STAFF REVIEW · H446 229 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. A particular instance of a class.

2. Initialises a new object.

3. Ends the function and passes a result to its caller.

4. A pure function has no externally observable side effects and returns the same result for the same inputs.

I Do — source worked steps

1. Predict — It is procedural when used as a standalone function in a procedural programme.

2. Model response — Square(3) returns 9; it does not change object state. C# can use more than one paradigm.

3. Follow the declared and actual types — Both variables are declared as BaseMessage . Their actual objects are EmailMessage and TextMessage , each inheriting from the base class.

4. Why is virtual important? — The base method is virtual and both derived classes override it. The language dispatches to the actual object's implementation.

5. Output and design consequence — Outputs Email then Text . The caller can add another derived message class without writing separate type-checking branches.

We Do — guided CHECK

1. Model response
This is object-oriented design; after Add(2), its Points becomes 7.

You Do — source mark schemes

1. [4] Procedural [1]; operations organised in subroutines [1]; parameters carry inputs [1]; functions can be reused/tested independently [1].

2. [6] Procedural code passes state to functions [2]; OOP stores account state with methods [2]; when a base method is virtual and the subclass overrides it, an Account reference to a SavingsAccount uses the SavingsAccount implementation [2].

3. [6] Applies a function to each element [2], returns transformed values [1], can avoid mutating original list [1]; C# also supports procedural code and mutable objects [1], so language/paradigm are not identical [1].

Exit Ticket — source answer

1. True. Programming paradigms overlap within multi-paradigm languages.